

Yuk Tung Samuel Fang

Also published as: **Yudong Fang**

Nanjing University (Suzhou Campus) — Hong Kong SAR — B.Sc. Intelligent Science and Technology (Exp. 2027)

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Research Interests

Robotics; perception and inference for embodied systems. Recent experience in navigation and semantic mapping; actively exploring broader directions. Seeking opportunities for overseas graduate studies (Fall 2027 entry).

Education

Nanjing University (Suzhou Campus)

Suzhou, China

B.Sc. Intelligent Science and Technology

Expected 2027

- **Total GPA:** 4.44 — **Core GPA:** 4.42 — **Sophomore GPA:** 4.543 — **Rank:** 47/151 (Top 31.2%)
- **Selected coursework (scores):** Database Systems (96), Operating Systems (95), Data Structures & Algorithms (95), Machine Learning (94), Deep Learning (94), Programming Practicum (94.9)

Research Experience

Inference & Learning Research Group (led by Prof. Yang Gao), Nanjing University

State Key Laboratory for Novel Software Technology

Undergraduate Researcher, 2025–present

- Supervised by tenure-track AP Prof. Jieqi Shi, who provided guidance on research design, paper writing, and project execution; collaborated closely on prototyping, integration, and manuscript preparation.
- Conducted research in embodied robotics, replicating experiments in navigation and exploration, while also replicating, integrating, and innovating in semantic mapping, with successful deployments on real robots.
- Reproduced and integrated state-of-the-art models (e.g., RAM, SAM3/2, DINOv3/2, Grounded DINO, FM-Fusion, EdgeTAM, VINS-Mono/Fusion, RAG, Clip) for semantic scene understanding and navigation; customized source code for end-to-end pipelines, including MLLM API calls and local deployments.
- Managed ROS bag streaming, real-time data monitoring/visualization, and hardware bring-up across diverse sensor/compute configurations (Jetson AGX Orin with 4× RealSense D435i, 4× Gemini 336L, Manifold Odin 1).
- Designed and 3D-printed custom mounting brackets; resolved engineering challenges such as power delivery, bandwidth constraints, port conflicts, and communication stacks (camera → compute → robot) to enable stable on-robot execution on platforms like Unitree Go1.

Publications & Preprints

- Ding, H.; Xu, Z.; **Fang, Y. (Yudong)**; Wu, Y.; Chen, Z.; Shi, J.; Huo, J.; Gao, Y.
LaViRA: Language-Vision-Robot Actions Translation for Zero-Shot Vision Language Navigation in Continuous Environments. *ICRA 2026*. arXiv:2510.19655 — Project Page
Contribution: Led real-robot deployment (on Unitree Go1) and system integration. (2025.05 – 2025.09)
- Ding, H.; Liang, X.; **Fang, Y. (Yudong)**; Wu, Y.; Shi, J.; Huo, J.; Li, W.; Wu, J.; Lai, Y.-K.; Gao, Y.
SEA: Semantic Map Prediction for Active Exploration of Uncertain Areas. *Under review*. arXiv:2510.19766 — Project Page
Contribution: Led real-robot deployment (on Unitree Go1) and end-to-end integration for on-robot execution. (2025.09 – 2025.10)
- **Fang, Y. T. S.**; Shi, Z.; Qiu, J.; Chen, Z.; Shi, J.; Xu, H.; Huo, J.; Gao, Y.
INHerit-SG: Incremental Hierarchical Semantic Scene Graphs with RAG-Style Retrieval.
Under review. arXiv:2602.12971 — Project Page
Role: Independent first author; led end-to-end pipeline, writing, figures, experiment design, visualization, supplement, hardware/communication bring-up, and real-robot data collection. (2025.10 – 2026.02)
Team contributions: Dataset construction, pipeline optimization, baseline adaptation and benchmarking, experiment table data collection, and human-study design/data.

Honors & Awards

- NJU Scholarship for HK/Macao & Overseas Chinese Students — **First Prize** (Sophomore year; university-wide; 5 awardees; defense required)
- NJU Scholarship for HK/Macao & Overseas Chinese Students — **Third Prize** (Freshman year; university-wide)

Skills

- **Programming:** Python, C++ — **Robotics:** ROS, real-robot deployment and debugging
- **Tools:** Linux, Git; proficient with AI-assisted coding tools for rapid prototyping/debugging; 3D modeling & printing